

UNIT II

Speech Recognition

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- Real and Complex Cepstrum
- Application of cepstral analysis to speech signal
- Feature extraction for speech
- Static and dynamic feature for speech recognition
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- Distance measures
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Topic

- Cepstrum Analysis
- Real and Complex Cepstrum
- Application of cepstral analysis to speech signal

Cepstral Analysis

- Cepstrum analysis is a nonlinear signal processing technique.
- The cepstrum technique performed on a time-domain signal to look at periodic content in data, such as harmonics, sidebands, and orders.
- The word cepstrum was coined by reversing the first syllable in the word spectrum.
- The cepstrum exists in a domain referred to as quefrency (reversal of the first syllable in frequency) which has units of time.

Cepstral Analysis

- It is calculated by first taking the Fourier transform of a time signal to create a frequency spectrum, converting it to a log scale, and then performing an inverse Fourier transform on the spectrum
- The cepstrum is referred to as a “spectrum of a spectrum” because it involves calculating the spectrum of the time signal and then performing another spectral operation.
- Because the cepstrum comes from an inversion of spectrum, the name “cepstrum” is derived by reversing the first few letters of “spectrum”.

Cepstral Analysis

- Many of the terms related to the cepstral domain have also been altered to indicate that cepstral processing has occurred.
- Some common ones include:
 - Quefreny from frequency
 - Saphe from phase
 - Liftering from filtering
 - Rahmonics from harmonics
- Because only the log of the amplitude is kept when the signal is in the frequency domain, it doesn't translate back to the same time signal when a inverse Fourier transform is performed. Instead, the signal is now in the "quefreny" domain.
- The cepstrum transform looks for periodic components in the frequency domain.
- As harmonic content in a frequency spectrum is evenly spaced, the cepstrum transform is particularly good at identifying harmonics.

Cepstral Analysis

- Steps to transform a time signal into the cepstral domain:

STEP 1:

Take the time signal and apply a FFT on the data to translate the time data into frequency data.

STEP 2:

Discard the phase portion of the FFT by converting the amplitude to a $\log_{10}$ of the data. This step reduces the dynamic range of the data by compressing it to a logarithm and makes it easier to find harmonic components.

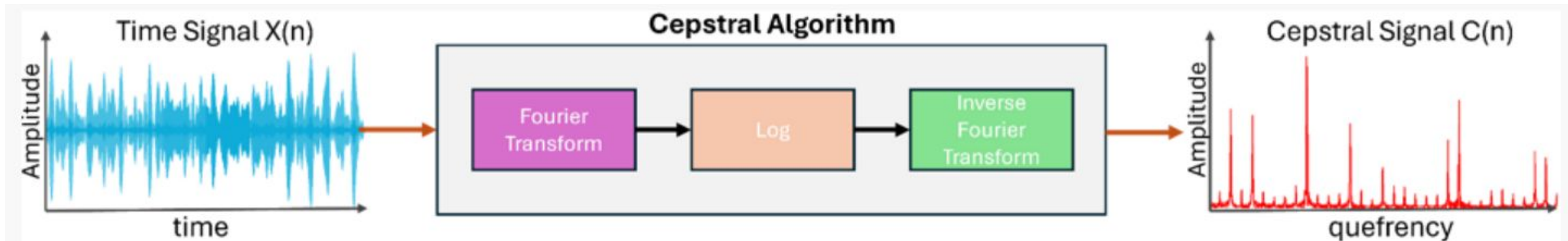
STEP 3:

Convert the frequency signal to the quefrency domain using an Inverse Fourier Transform. The phasing portion of an Inverse Fourier Transform rotates in the opposite direction of a Fourier Transform. Since only the amplitude was kept and the log of the amplitude was applied, it converts to the quefrency domain.

Cepstral Analysis

The mathematical representation of cepstral analysis

$$C[n] = \sum_{n=0}^{N-1} \log \left(\left| \sum_{n=0}^{N-1} X[n] e^{-j\left(\frac{2\pi}{N}\right)kn} \right| \right) e^{j\left(\frac{2\pi}{N}\right)kn}$$

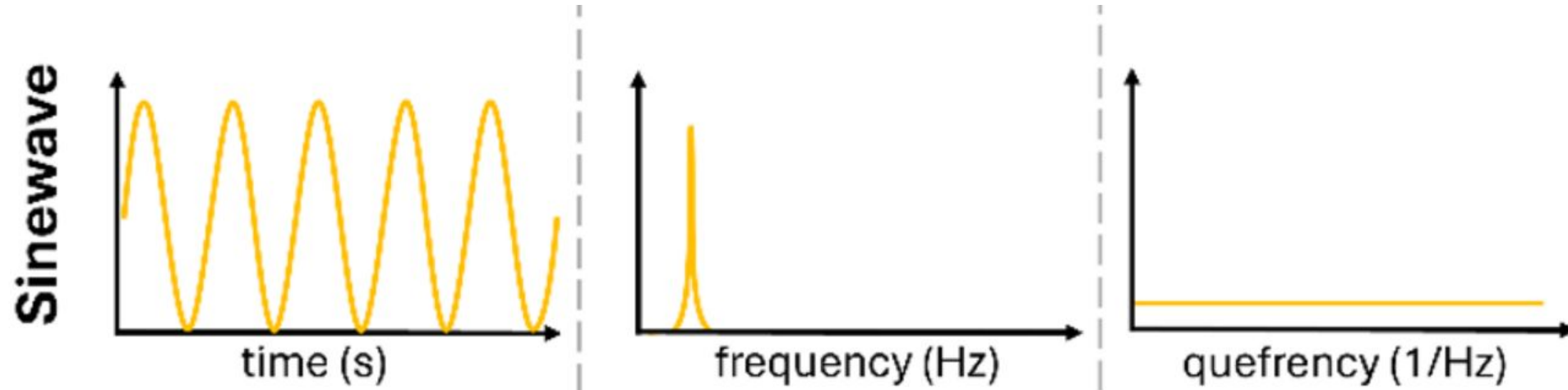


Cepstral Analysis

- The convolution of two signals (Gx and Fx) in the time domain, is equal to multiplying the signals in the frequency domain.
- Due to the logarithmic nature of the cepstrum, this is equal to the addition of two signals ($Gx + Fx$) in the cepstral domain.
- This can make cepstrum useful for separating signals in a mathematically more efficient way.
- This is commonly used in voice analysis for determining a voice against a background.
- This principle is how a voice track can be separated from a music track in karaoke for example.
- Cepstral analysis is used in speech and image processing.

Cepstral Analysis

- Representation of sine wave in different domain:



- Two components of cepstral analysis:
 1. Real Cepstrum
 2. Complex Cepstrum

Real Cepstrum

- The real cepstrum of a signal $x(n)$, sometimes called simply the cepstrum.
- It is calculated by determining the natural logarithm of magnitude of the Fourier transform of $x(n)$, then obtaining the inverse Fourier transform of the resulting sequence.
- Real cepstrum of a digital signal $x(n)$ is defined as:
- The Real Cepstrum $c(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \ln|X(\omega)| e^{j\omega n} d\omega$ is the natural logarithm of the magnitude of the signal's spectrum, discarding phase information.

Real Cepstrum

- Steps to calculate Real Cepstrum: $(C_r(n))$
 - I. Compute the DFT (F) of the signal $x(n)$.
 - II. Take the absolute magnitude of the result.
 - III. Compute the natural logarithm of that magnitude.
 - IV. Take the Inverse DFT (IDFT)

$$c_r(n) = \mathcal{F}^{-1} \{ \ln | \mathcal{F} \{ x(n) \} | \}$$

- Because the real cepstrum discards phase data, it is used when you only need to analyze the periodicities or the overall shape (envelope) of a signal.

Real Cepstrum

Applications of Real Cepstrum:

- Speech Analysis:

- It extracts the pitch (fundamental frequency) of a speaker.
- It also helps isolate the spectral envelope, which is used in speech recognition to identify vowels and consonants.

- Audio Processing:

- It is the foundation for creating Mel-Frequency Cepstral Coefficients (MFCCs), the core feature used by voice assistants (like Alexa or Siri) to understand commands

Complex Cepstrum

- The complex cepstrum of a sequence $x(n)$ is calculated by finding the complex natural logarithm of the Fourier transform of $x(n)$, then the inverse Fourier transform of the resulting sequence.

- The complex cepstrum is defined as:

$$\hat{x}(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \ln X(\omega) e^{j\omega n} d\omega$$

- The Complex Cepstrum uses both the magnitude and the phase, making it a fully reversible transformation.

Complex Cepstrum

- Steps to calculate Complex Cepstrum:

I. Compute the DFT of the signal $x(n)$.

II. Express the complex result in polar form.

$$X(\omega) = |X(\omega)|e^{j\angle X(\omega)}$$

III. Compute the complex logarithm.

$$\ln(X(\omega)) = \ln|X(\omega)| + j\angle X(\omega)$$

IV. Take the Inverse DFT (IDFT)

$$c_c(n) = \mathcal{F}^{-1} \{ \ln(\mathcal{F}\{x(n)\}) \}$$

- Because the complex cepstrum preserves phase data, it is used when you need to modify the signal and convert it back into a clean, usable time-domain signal.

Complex Cepstrum

Applications of Complex Cepstrum:

- Echo Removal & Dereverberation:
 - It identifies the exact delay time of an echo. By removing the echo peak in the "quefreny" domain and converting it back, it completely removes echoes from audio or telecom recordings.
- Speech Synthesis (Text-to-Speech):
 - It allows engineers to separate the vocal tract filter from the glottis source. Designers can then modify the pitch or duration independently to create natural-sounding robotic speech.
- Deconvolution of Blurring:
 - It restores signals or one-dimensional imaging data that have been distorted or blurred by a transmission channel.

Cepstral analysis for speech signal

- Cepstral analysis is primarily used in speech processing to separate the excitation source (pitch) from the vocal tract filter (formants).
- This mathematical process deconvolves a speech signal by transforming a multiplicative relationship in the frequency domain into an additive one in the cepstral domain.
- Speech $s(n)$ is generated by convolving an excitation source $e(n)$ (glottal pulses or air noise) with a vocal tract filter $h(n)$:

$$s(n) = e(n) * h(n)$$

- In the standard frequency domain, this convolution becomes a multiplication:

$$|S(w)| = |E(w)| |H(w)|$$

Cepstral analysis for speech signal

- Linear filters cannot easily separate multiplied signals.
- Logarithm transforms takes the mathematical relationship from multiplicative to additive:

$$\log |S(w)| = \log |E(w)| + \log |H(w)|$$

where $\log |H(w)|$ = Spectral envelope (vocal tract shape)

$\log |E(w)|$ = rapidly varying ripples (harmonics/pitch).

- Because the vocal tract changes slowly, its log-spectrum is smooth (low frequency).
- Because the glottal excitation consists of periodic pulses, its log-spectrum has sharp, rapid ripples (high frequency).

Cepstral analysis for speech signal

- Real Cepstrum of a speech signal

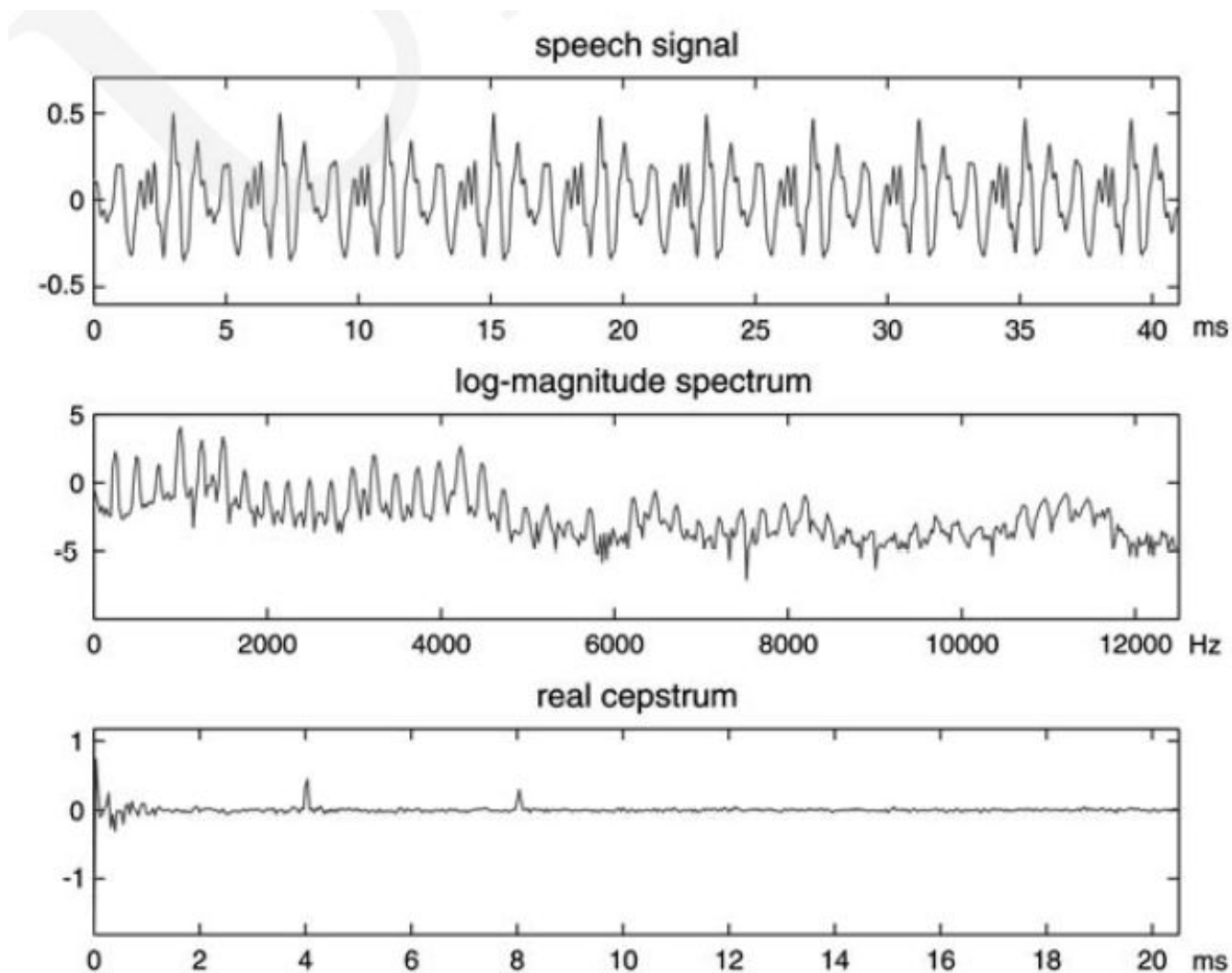


Fig. 1. A segment of speech signal, log-magnitude spectrum and cepstrum